Physical Diffusion

Assume that this process can be model with a Gaussian distribution

Flow Matching

- Forward diffusion simulation free in a flow matching context
- We have a closed form for getting from clean image to noise image at time step t

High level diagram

- Reduces compute overhead

VAE Architecture

- Encoder/decoder are symmetric in the spatial up and down sampling

Residual Blocks

- This approach keeps the gradient clean, whereas post activation that was common in early residual networks, but still causes gradient vanishing in deep models because the shortcut is behind an activation and norm.

Attention Blocks

- The speedup of flash attention 2 is in memory management

Spatial Resizing

- Bilinear and bicubic interpolation introduces blurring that needs to be discarded by the VAE
- The simpler approach produces less complex artifacts

FSQ Layer

- No need for loss adjustments to utilise the whole codebook

Why $2\sigma(1.6) - 1$?

- Quite clsoe to uniform distribution, making codebook utilisation rise faster

VAE Summary

- All of this is to make the DiT's job easier

Training the VAE: Loss

- L1 handles structure
- Adversarial handles high frequency details like fur and skin

Training the VAE: Warmup and Cooldown

- The discriminator can't be too strong too early

What's been skipped?

- Spectral norm keeps the discriminator's gradients from exploding and feeding back to the VAE

Attention

- RoPE is virtually required because the transform is position blind, whereas RoPE gives an ordering on neighbouring patches

Adaptive Layer Norm

- Initialised to 0, which allows the model to start as the identity function, preventing gradient explosions and destablisations

Training the DiT-UNet

- Flow Matching loss is just MSE between the predicted velocity and actual velocity

Optimisations and Stability

- EMA is what is actually used to do the sampling
- float32 is required for the loss due to overflow issues

We've got a model, now what? RK2

- Take an Euler half step, then evaluate from that step where the velocity should be, then go back to the original step and move in the direction predicted by that half step
- Alternative you can make a full step, evaluate, then go back and use the average of the full steps velocity, and the original points velocity (heun)

CFG

- The purpose is to align the model with the text prompt more

Post Sampling

- Snapping is optional, but tends to help the VAE produce non blurry results

Mid-Training

- Loss appears plateaued, but images get better throughout this plateau, without affecting the quality of other images